

Root Cause Analysis Report

MLOps House Price Prediction | Deployment Incidents

Period: 29-30 September 2026

Prepared by

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1. Executive Summary

The MLOps house-price prediction system trains a RandomForestRegressor on cleaned data from Amazon S3, tracks it in MLflow 3.16.1, registers it as **house_price_prediction** and serves it through a Dockerised FastAPI application on AWS EC2. Between 29 and 30 September 2026, eight deployment incidents were identified and analysed. All were traced to configuration, dependency or connectivity gaps rather than model defects. The final state is healthy: model version 1 (alias *champion*) is READY and the FastAPI container is running on port 8000.

2. System Overview

Data / Model	S3 cleaned CSV (980 rows, 6 features); RandomForestRegressor (max_depth=8, n_estimators=150, random_state=42)
Serving	FastAPI + Uvicorn on port 8000, custom HTML/CSS UI; loads models:/house_price_prediction@champion
Platform	MLflow tracking / registry on port 5000, Docker, AWS EC2, GitHub Actions

3. Incident Register and Root Causes

ID	Symptom	Root Cause	Resolution
INC-001	MLflow connection refused (port 5000)	MLflow server installed but not running; no startup or readiness check.	Started MLflow on EC2, verified listener and API with curl, restarted FastAPI.
INC-002	AWS NoCredentials on EC2	No usable credentials or IAM role for the AWS CLI / application.	Installed AWS CLI, granted EC2 IAM access, validated with aws s3 ls.
INC-003	Docker build failure	Python 3.11 incompatibility; Windows-only pywin32==312 in Linux requirements.	Moved to python:3.12-slim; removed OS-specific packages.
INC-004	skops untrusted type during model logging	skops trust validation rejected sklearn.tree._tree.Tree (serialization issue, not training).	Adjusted serialization path; verified model loads in target environment.
INC-005	S3 operation failure	Trailing space in bucket-name configuration.	Normalised configuration and removed whitespace.
INC-006	404 for champion model	Model name mismatch (house-price-predictor vs house_price_prediction) or missing alias.	Standardised on house_price_prediction; verified v1 and champion alias.
INC-007	FastAPI to MLflow connection refused	MLflow unreachable from the container (not listening / wrong host).	Kept MLflow reachable on 5000; fixed container networking and MLFLOW_TRACKING.
INC-008	ERR_SSL_PROTOCOL_ERROR	Plain-HTTP MLflow accessed over HTTPS.	Use http://<EC2-IP>:5000; add TLS proxy only if HTTPS is required.

4. Common Root-Cause Themes

- **Missing dependency readiness:** the application relied on MLflow and S3 without startup or health validation (INC-001, 002, 007).
- **Environment drift:** Windows-only packages and Python version mismatches entered the Linux image (INC-003).
- **Configuration inconsistency:** model names, aliases, bucket names and URL protocols were not standardised or validated (INC-005, 006, 008).
- **Serialization compatibility:** model logging depended on library-version trust rules (INC-004).

5. Corrective and Preventive Actions

- Run MLflow as a managed service (systemd/Docker) with restart policy, readiness check and port-5000 monitoring.
- Use an EC2 IAM role with least-privilege S3 access; validate access at startup. Never commit .env files or credentials.

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- Maintain a Linux-only production requirements file; pin Python, MLflow and scikit-learn versions; build the image in Linux CI.
- Define MODEL_NAME and MODEL_ALIAS once as environment variables and validate them at application startup.
- Add a CI test covering train, log, register, load, predict and package before every deployment.
- Use immutable image tags, TLS via reverse proxy if exposed beyond a trusted network, and document rollback (previous image tag + model version).

6. Final State and Validation

MLflow shows **house_price_prediction** version 1 with the *champion* alias in READY state. The FastAPI container *house-price-prediction* runs on port 8000 with registry-based model loading, and MLflow is consumed over HTTP on port 5000. Recommended sign-off checks: MLflow reachable, S3 accessible from the runtime, alias resolves, `/health` healthy, `/docs` and UI load, and `/predict` returns a valid prediction.

Note: INC-002 and INC-004 require validation of the final IAM and serialization configuration; exact commands and infrastructure details not captured in the source documentation are intentionally omitted.